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Host filtering and microbial niche breadth organize the global seed microbiome

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Order of Authors:	Zilong Wang
	Wenfeng Ai
	Xiangyu Li
	Ying Xu
	Peng He
	Tengfei Ma
	Ning Ling
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Host filtering and microbial niche breadth organize the global seed microbiome

Authors:

Zilong Wang [1], Wenfeng Ai[1], Xiangyu Li[1], Ying Xu[1], Peng He[1], Tengfei Ma [1], Ning Ling [1, 2, *]

Affiliations:

[1] *State Key Laboratory of Herbage Improvement and Grassland Agro-ecosystems, College of Pastoral Agriculture Science and Technology, Centre for Grassland Microbiome, Lanzhou University, Lanzhou, 730000, Gansu, China.*

[2] *Jiangsu Provincial Key Lab for Organic Solid Waste Utilization, Jiangsu Collaborative Innovation Center for Solid Organic Waste Resource Utilization, Nanjing Agricultural University, Nanjing, 210095, China.*

*** Corresponding authors:**

Ning Ling, *E-mail: nling@njau.edu.cn*

Summary

Seeds provide a critical interface for microbial transmission across plant generations, yet how host filtering, phylogenetic relatedness, geography, and microbial niche breadth jointly structure seed microbiomes remains unclear at the global scale. Here, we integrated 6,474 seed-associated microbial samples from 176 plant species and 123 genera across 46 countries, combining publicly available datasets with newly generated seed microbiome data, to uncover the ecological processes governing global seed microbiome assembly. Our analysis reveals that host taxonomic identity is the dominant driver of global seed microbiome assembly, exerting a stronger influence than geographic origin. Remarkably, ASV-level profiles outperformed genus-level profiles in predicting host identity, providing a robust and reproducible microbial signature of host identity. Furthermore, we uncovered a systematic divergence in microbial ecological strategies. Bacterial generalists showed greater physiological tolerance, metabolic versatility, and genomic capacity, with traits linked to biofilm formation, resource acquisition, and stress adaptation, whereas specialists were enriched in anaerobic and thermotolerant traits. Together, these analyses reveal how host identity and microbial ecological strategies jointly shape the assembly and functional organization of the global seed microbiome, providing a unified framework for understanding host–microbe interactions across plant lineages.

Key word: Seed microbiome, Host filtering, Host–microbe interactions, Microbial assembly, Generalists and specialists, Phylosymbiosis

Introduction

As the fundamental units of plant reproduction and dispersal, seeds carry not only the genetic information of their hosts but also vertically transmitted microbial communities shaped by a stringent transmission bottleneck [1-3]. These seed-borne communities support germination, seedling establishment, nutrient acquisition, and overall plant fitness [4-6]. The assembly of the seed microbiome reflects the combined influence of host-associated and spatial processes, with host genetic variation exerting strong deterministic effects on community composition [7,8], while geographic origin, edaphic properties, and climatic conditions contribute additional spatial structure [9-11]. Yet, most studies have focused on a few species, a limited number of sites, or individual factors, leaving unresolved how these processes jointly shape seed microbiome assembly at the global scale [12].

Host identity is a major determinant of microbiome assembly, with phylogenetically similar hosts tending to harbor more similar microbial communities, even under different environmental conditions [7,13]. Such phylosymbiotic patterns have been observed in root, phyllosphere, and endophytic communities [14-17] , potentially reflecting conserved host traits, including immunity, nutrient acquisition, and metabolite production, that shape microbial recruitment and persistence [7,18]. Seeds provide a particularly informative context for testing this relationship because their microbiota is assembled within a reproductive structure and constrained by a stringent transmission bottleneck [3,6]. These distinctive assembly processes may strengthen host-mediated filtering, yet whether host relatedness broadly organizes seed microbiome assembly across plant lineages remains unresolved.

The host-associated structure of the seed microbiome suggests that strong and reproducible host filtering should leave a detectable signature of host identity in microbial community composition, extending the host phenotype into its associated microbial community [19,20]. Consistent with this premise, microbial fingerprints have been used to distinguish seed origins and, in some systems, host genetic backgrounds, demonstrating that seed-associated communities can retain information linked to their plant hosts [10,21-23]. The stability and vertical transmission of seed microbiota may further preserve these host-associated signals across

generations [24,25]. However, whether host identity itself can be reliably recovered from seed microbial community composition remains largely unexplored, particularly across diverse host taxa.

Host filtering raises a fundamental question: how do microorganisms differ in their responses to the host environments they encounter? Microorganisms vary markedly in host breadth, ranging from narrow-niche specialists to broad-niche generalists, representing distinct ecological strategies for coping with host-associated selection [26-28]. Plant-associated communities encompass both taxa with consistent associations with particular hosts or organs [29] and taxa capable of persisting across diverse hosts and environmental contexts [30]. Generalists typically tolerate broader environmental conditions and can exploit diverse resources, traits that promote persistence across heterogeneous habitats and under disturbance [31-33]. Specialists, in contrast, occupy narrower niches but may achieve high fitness under particular environmental or host conditions [27,31]. These strategies are not simply opposite levels of ecological success: generalism is often associated with broader functional repertoires and higher abundance, whereas specialization can arise through diverse ecological and evolutionary routes and remain locally advantageous [27,28,34]. Such differences in niche breadth may shape how microorganisms respond to host filtering and persist across plant lineages. Yet whether generalists and specialists differ systematically in their ecological prevalence and functional strategies within the seed microbiome remains unknown.

Despite growing evidence that host identity shapes seed microbiomes [35,36], how host evolutionary history and microbial niche breadth influence these associations remains unclear. Using a global cross-host dataset, we tested how host identity and evolutionary relatedness structure seed microbiomes and whether microbial prevalence across hosts reflects distinct niche strategies. We traced the effects of host filtering across multiple levels, asking how it shapes community assembly and host phylogenetic structure, whether microbial communities retain signatures of host identity, and how microbial ecological and functional strategies differ with host niche breadth. We hypothesized that seeds act as strong host-filtering interfaces, generating reproducible host-associated microbial signatures while favoring distinct microbial strategies along the continuum of host breadth. Together, these analyses provide an integrated

framework for understanding how host identity and microbial ecological strategies jointly organize the global seed microbiome.

Results

A global seed microbiome dataset

To define how host identity, geography, and evolutionary relatedness shape seed-associated microbiomes, we assembled a cross-study amplicon dataset integrating publicly available data (S1–S58; Table S1), a published meta-analysis (S59), and our own 16S and ITS sequencing (S60). The dataset comprised 6,474 samples from 59 studies, 46 countries, and 176 host species spanning 123 genera, including 4,107 bacterial communities profiled by 16S and 2,367 fungal communities profiled by ITS across six amplicon regions (Fig. 1C, D; Table S2). Literature-derived datasets contributed most samples, with V4 and ITS1 the predominant regions (Fig. 1C, D). We harmonised sequences across marker regions using reference sequences to generate a comparable feature matrix.

Sampling was global but concentrated in Western Europe, Oceania, East Asia, and North America, with France (1,593), Australia (1,069), China (1,024), and the United States (736) together contributing 68.7% of samples (Fig. 1A, B). Most samples were mature seeds (74.1%), while the remainder were seedlings germinated aseptically from surface-sterilised seeds (Fig. 1G). Both represented intra-seed microbiomes, and samples from other plant compartments or with unclear host identity were excluded. Host sampling was concentrated in crop genera, with *Brassica*, *Oryza*, *Glycine*, and *Triticum* dominating 16S datasets and *Brassica*, *Oryza*, *Raphanus*, and *Phaseolus* dominating ITS datasets (Fig. 1H, I; Table S3).

We retained ASVs that failed to map to reference databases: 65.6% of 16S and 58.2% of ITS ASVs lacked matches at ≥97% identity and were assigned secondary taxonomy using sintax, preserving potentially underrepresented seed-associated taxa. Taxonomic annotation was substantially deeper for bacteria than fungi, with median genus-level assignment of 93.3% of 16S reads versus 44.9% of ITS reads (Fig. 1E, F), likely reflecting poorer reference coverage of seed-associated fungi; this difference was accounted for in cross-kingdom comparisons. Together, the dataset provides broad

phylogenetic, geographic, and taxonomic coverage for testing how host identity shapes seed microbiome assembly.

Host identity governs seed microbiome assembly

Having established the dataset, we first asked whether host identity shapes seed microbial diversity. Host identity strongly structured seed microbial α-diversity across all three datasets (16S V4, 16S V5–V7, ITS1) and taxonomic levels (Kruskal–Wallis, $p < 1.5 \times 10^{-32}$; Fig. S1, S2), indicating that host identity consistently structures seed microbial α diversity in both bacteria and fungi.

We next assessed whether this host effect is evolutionarily conserved using Pagel's λ on the host species phylogeny. Bacterial α diversity showed strong phylogenetic signal across both 16S regions (λ = 0.76–0.87, all $p < 0.05$), indicating greater similarity in diversity among closely related hosts (Fig. 2A, B, D, E), whereas fungal α diversity showed essentially no signal (λ ≈ 0, $p = 1$; Fig. 2C, F). This contrasting pattern was robust to within-species sampling uncertainty, as bootstrap resampling consistently supported high λ values for bacterial diversity and little evidence of phylogenetic signal for fungal diversity (Fig. S3). Thus, bacterial α diversity is more closely coupled to host evolutionary history than fungal α diversity.

This phylogenetic association also extended to community composition, but with varying strength across datasets. Host phylogenetic distance was significantly correlated with species-level mean Bray–Curtis dissimilarity in all three datasets (Mantel $p \leq 0.05$), with the strongest signal in V4 bacteria (Mantel r = 0.471; MRM R^2 = 0.222), an intermediate signal in ITS1 fungi (r = 0.221; R^2 = 0.049), and the weakest in V5–V7 bacteria (r = 0.078; R^2 = 0.006) (Fig. 2G–I). Notably, fungi retained a host-phylogeny dependence at the community level despite lacking a phylogenetic signal in α diversity, suggesting that host specificity need not closely track host phylogeny.

To assess the relative contributions of host and technical factors, we partitioned β diversity by host genus, country, and DNA-extraction method. Host genus consistently emerged as the strongest predictor across V4, V5–V7, and ITS1 communities (R^2 =

0.118, 0.353, and 0.336; all $p < 0.001$), exceeding country and extraction effects and contributing the largest unique fractions of variance (10.4%, 30.9%, and 31.0%; Fig. S4). This result remained consistent after collapsing ASVs to microbial genera and pooling amplicon regions, with host genus still the strongest factor for both bacteria and fungi (marginal $R^2 = 0.210$ and 0.226, respectively; $p < 0.001$; Fig. S5). Host genus therefore emerged as the most reproducible taxonomic scale for explaining seed microbiome composition.

Host and geographic effects were further reflected in the processes underlying community assembly. In the laboratory-controlled S60 dataset, geographic and technical variation was minimized, allowing host-associated assembly processes to be examined more directly. With increasing divergence among host lineages, community assembly progressively shifted from drift and homogenising dispersal toward dispersal limitation and heterogeneous selection. This transition became increasingly pronounced for comparisons spanning host genera, orders, and classes, with heterogeneous selection particularly evident at broader phylogenetic contrasts (Fig. 2J). Within host-genus pools, geographic separation produced a different signature. In the V4 dataset, comparisons across countries showed a marked reduction in drift, from 63% to 41%, and a corresponding increase in dispersal limitation, from 19% to 40%, together with smaller increases in heterogeneous and homogeneous selection (Fig. 2K, L). The overall composition of assembly processes differed significantly between same- and cross-country comparisons (Cramer's V = 0.175, Monte Carlo $p < 0.001$; Fig. 2K). These results distinguish two complementary dimensions of seed microbiome assembly: increasing host divergence is associated with stronger deterministic selection, whereas geographic separation within the same host-genus pool primarily enhances dispersal limitation.

The seed microbiome as a predictable host phenotype

The dominance of host identity over community composition raises a sharper question: can host identity itself be recovered from the microbiome? To test this, we

subjected each dataset`s between-sample Bray – Curtis dissimilarity matrix (no similarity truncation, T = 0) to unsupervised hierarchical clustering across a full grid of eight linkage methods and k = 2–400 clusters, and scored each partition against host labels at four taxonomic levels (order, family, genus, species) using the adjusted Rand index (ARI), which measures agreement corrected for chance (0 = random, 1 = perfect; Fig. 3A–F). At the host-genus level, clustering reached an ARI far above chance in all three datasets, 0.421 for V4 (n = 2,450), 0.638 for V5-V7 (n = 745), and 0.663 for ITS1 (n = 1,413; all with mcquitty linkage), and the optimal cluster numbers (k = 59, 83, 154) were of the same order as the number of host genera in each dataset (58, 64, 85). Peaks at the order, family, and species levels were comparably strong (0.429/0.429/0.428 for V4; 0.365/0.377/0.638 for V5-V7; 0.571/0.579/0.651 for ITS1). Notably, a purely internal clustering criterion, the silhouette coefficient, recovered host genus far less well (ARI 0.117, 0.202, and 0.435), underscoring that host-associated structure captures an axis not fully reflected by internal cluster compactness (Fig. 3A – C). Community composition could classify seed samples by host genus with considerable fidelity, with the strongest signal observed for fungi (ITS1) despite their near-zero α-diversity phylogenetic signal (Fig. 2C, F). Thus, strong host-genus specificity in microbial composition can occur independently of host phylogenetic relatedness (Fig. 3C, F).

This predictability depends critically on the taxonomic resolution of the microbial features. When ASVs were collapsed to microbial genera and all amplicon regions pooled into single 16S (n = 4,082) and ITS (n = 2,352) tables, and the identical clustering and ARI protocol was repeated, host-genus-level ARI collapsed to less than half its species-level value, 0.189 for 16S and 0.212 for ITS (both ward.D2), with peaks at every other host level falling in tandem (species/order/family: 0.213/0.129/0.132 for 16S; 0.217/0.128/0.125 for ITS), such that no 16S linkage method reached an ARI of even 0.20 (Fig. S6A–D). The information that allows host identity to be recovered is therefore carried substantially at the subgenus (species/strain) level, within microbial genera, rather than by genus-level composition alone. The seed microbiome therefore

carries a reproducible host-associated signature, a signature concentrated in fine-scale microbial sequence variants that are lost upon genus-level aggregation.

Host filtering raises a further question: how do microorganisms differ in the breadth of hosts they can colonize? Microorganisms may range from narrow-niche specialists restricted to specific hosts to broad-niche generalists capable of spreading across diverse hosts, reflecting distinct ecological responses to host-associated selection. We therefore developed a rigorous two-stage framework to define generalists and specialists at the fine-scale sequence-variant level.

Generalists and specialists exhibit systematic taxonomic and functional divergence

Generalists and specialists were distinguished by host breadth and phylogenetic dispersion, revealing a pronounced but asymmetric ecological divide between the two groups (Fig. S7). ASVs detected in at least five host species were first identified as candidates, after which lineage-restricted ASVs associated with phylogenetically clustered hosts were excluded to avoid classifying host-specific lineages as generalists. This removed 32 bacterial and 41 fungal ASVs, particularly those associated with *Oryza*, *Brassica*, and *Triticum* (Fig. S7; Table S4). The final classification comprised 1,103 bacterial and 235 fungal generalists and 5,830 bacterial and 1,232 fungal specialists, with specialists defined as ASVs restricted to a single host species (Table S5).

The distinction was strongest in bacteria, where generalists were concentrated in Gammaproteobacteria, Alphaproteobacteria, Bacilli, and Actinobacteria, which together accounted for 93.9% of generalists. The ten most common genera represented 45.8% of bacterial generalists and were led by *Pseudomonas*, *Massilia*, *Paenibacillus*, *Enterobacter*, and *Pantoea*. Generalists were predominantly aerobic or facultatively anaerobic (95.9%), biofilm-forming (83.9%), and Gram-negative (75.3%), whereas specialists spanned a broader phylogenetic range and contained a substantially larger anaerobic fraction (32.3% versus 4.1%) (Fig. 4A, B). Consistent with this pattern, 13

genera, including *Pseudomonas*, *Pantoea*, *Enterobacter*, *Massilia*, and *Paenibacillus*, were significantly enriched among generalists, whereas *Lachnospiraceae* NK4A136, *Blautia*, and *Bacteroides* were specialist-enriched (Fig. S8A, B). In fungi, by contrast, taxonomic separation was much weaker: only *Alternaria* was enriched among generalists and Agaricomycetes among specialists, while dominant fungal taxa occurred at broadly similar proportions between the two groups (Fig. S8C, D).

This taxonomic contrast was closely linked to prevalence and abundance. All 13 generalist-enriched bacterial genera ranked among the 25 most widespread genera, while *Pseudomonas* and *Pantoea* alone contributed 28.1% of mean bacterial relative abundance; the top 25 genera accounted for 50.9% overall. The corresponding proportion in fungi was 25.2% (Fig. 4C, D). Across dominant bacterial and fungal lineages, ASV abundance increased with host breadth (all $p < 0.001$), placing generalists consistently at the high-abundance end of the distribution (Fig. S9). Thus, particularly in bacteria, broad host distribution was associated with a concentrated set of widespread and abundant taxa rather than simply greater phylogenetic diversity.

Having established this discrete contrast, we next asked whether it reflected a broader functional gradient by relating predicted traits directly to the continuous host-breadth measure, *n_*hosts, defined as the number of host species in which each ASV was detected (Fig. 5). In bacteria, 12 of 15 continuous metaTraits were significantly associated with host breadth, including lower growth-temperature thresholds (ρ = −0.208 to −0.140), larger genomes and higher gene counts (ρ = +0.157 to +0.165), higher GC content (ρ = +0.130), and broader pH and salinity tolerance ranges; 71 of 108 Boolean traits were also associated with *n_hosts* (Fig. 5A, B; Table S6). Generalists were enriched in stress tolerance, biofilm formation, aerotolerance, catalase activity, motility, facultative anaerobiosis, nitrate reduction, and diverse carbon-substrate utilization, whereas specialists were enriched in obligate anaerobiosis, Gram-positive cell walls, and growth at 42 °C. In fungi, only the Pathotroph guild increased with host breadth (ρ = +0.086, OR = 1.23, FDR = 0.036; Fig. 5B; Table S6).

The same gradient was evident in predicted gene repertoires. Of 9,580 bacterial

KOs, 5,623 (58.7%) were associated with *n_hosts*, with 2,781 positively and 746 negatively associated at |log2 OR| ≥ 0.5 per host-breadth doubling (Fig. 5C–E). Generalist-enriched KOs were dominated by transporters, transcription factors, two-component systems, and secretion systems, whereas specialist-enriched KOs were more functionally dispersed. Functional annotation similarly showed greater representation of signalling, genetic information processing, metabolism, and PGPG functions among generalist-enriched KOs, including substrate utilization, stress neutralization, phosphate solubilization, fitness, quorum sensing/biofilm formation, and motility/chemotaxis; specialist-enriched KOs were dominated by substrate utilization and fitness, with spore production showing the strongest enrichment (OR = 37.4, FDR = 2×10^{-15}) (Fig. S10). Thus, increasing host breadth was accompanied by larger, transport- and regulation-rich bacterial genomes and a broader predicted functional repertoire, rather than a universally superior competitive phenotype.

The particularly strong host restriction of some functional groups prompted us to examine four representative categories in greater detail: anaerobic and minimal-genome bacteria, which captured prominent associations with *n_hosts*, and mycorrhizal fungi and nitrogen-fixing bacteria, representing ecologically important host-associated functions (Fig. S11–S15). Across these groups, taxa were predominantly specialist or intermediate-breadth, occurred at low relative abundance, and showed strong partitioning among host genera (Fig. S11). Anaerobic taxa were particularly enriched in *Gossypium*, anaerobic and mycorrhizal taxa in *Achnatherum*, and anaerobic and nitrogen-fixing taxa in *Brassica*, while nitrogen-fixing bacteria were also prominent in *Oryza* (Fig. S11G–J); genus-level distributions showed similarly patchy host associations (Figs. S12–S15). Mycorrhizal fungi showed the strongest separation from generalists: although their association with *n*hosts was not statistically significant, their OR was only 0.13 (Table S6), and no generalist ASV was annotated as mycorrhizal, whereas mycorrhizal ASVs occurred among both specialists and Random taxa, the latter showing unstable occurrence across host species (Table S5). Nitrogen-fixing bacteria were the main exception, with 5.91% classified as generalists and 25.91% as

intermediate, such that 31.8% spanned broad host ranges, compared with 15.2% of anaerobic bacteria and 1.3% of mycorrhizal fungi (Fig. S11C–F). These broad-host-range diazotrophs were concentrated in *Azospirillum* and *Herbaspirillum*. Together, these patterns show that host breadth linked microbial identity and genomic function along a continuous gradient, while ecologically distinctive functional groups remained strongly partitioned among a limited set of host-associated reservoirs.

Discussion

Whether seed microbiome assembly is driven primarily by host identity, geography, or host relatedness has remained unresolved at the global scale. We substantially expanded previous coverage by assembling a global dataset of 6,474 seed microbiome samples from 60 studies, spanning 176 host species, 123 genera, and 46 countries. Despite methodological heterogeneity, major patterns were consistent across regions and microbial kingdoms, and variance partitioning accounting for extraction method still identified host identity as the dominant driver. Host genus was the strongest determinant of microbiome composition, and community composition reliably predicted host identity, particularly when fine-scale sequence variants were retained. Microbes were further partitioned into cross-host generalists and host-associated specialists, with anaerobic and mycorrhizal guilds showing strong host specialization and forming patchy distributions across host genera, whereas nitrogen-fixing bacteria spanned a substantially broader range of hosts, revealing a mosaic of host-associated functional niches. Together, these findings reveal a predictable, host-associated seed microbiome shaped by strong host filtering and contrasting microbial life-history strategies.

Phylosymbiosis, in which similarities in host-associated microbiomes mirror host phylogenetic relatedness, is increasingly recognized as a key driver of assembly in plant-associated microbiomes [17]. Our synthesis extends this framework to seeds and identifies host genus as the most reproducible taxonomic scale of assembly. Across all three amplicon regions, host genus was the strongest determinant of microbial β-diversity (marginal R^2 = 0.118–0.353), exceeding source country (0.017–0.062) and extraction method (0.002–0.009) (Fig. S4). Although the relative contributions of host and environment vary among plant microbiome systems, with soil type dominating some root communities and host phylogeny dominating others [13,37], these factors can act concurrently. Leaf-associated phylosymbiosis also varies among microbial groups [38]. In this context, the consistently strong host-genus effect in seeds supports host filtering as a major organizing force of the seed microbiome across diverse

environments and sampling conditions.

The relationship between host phylogeny and seed microbiomes was strongly kingdom dependent. Bacterial α-diversity showed substantial phylogenetic conservatism (Pagel's $\lambda = 0.76–0.87$), whereas fungal α-diversity showed virtually no signal ($\lambda \approx 0$) (Fig. 2). Yet fungi exhibited the strongest host-identity predictability (ARI = 0.663) (Fig. 3F) and remained correlated with host phylogenetic distance (Mantel r = 0.221) (Fig. 2I), indicating that fungal host association is expressed primarily through community turnover rather than diversity. This contrast is consistent with evidence from *Oryza*, where bacterial communities exhibited significant phylosymbiosis with host species, whereas fungal communities did not [38]. More broadly, fungal communities can be more strongly shaped by geographic and soil environmental variation than by host identity [39]. Thus, bacterial communities appear to retain a stronger phylogenetic imprint of hosts, whereas fungal communities reflect a more context-dependent combination of host-specific and spatial filtering [17].

Consistent with previous studies showing that spatial distance can strengthen dispersal limitation and alter the balance between stochastic and deterministic assembly processes [40-42], we found that geographic distance reshaped seed microbiome assembly among samples from the same host genus. Across countries, drift decreased from 63% to 41%, whereas dispersal limitation increased from 19% to 40% (Fig. 2L), indicating that increasing spatial distance progressively restricted microbial exchange among otherwise similar host-associated communities and weakened the homogenizing effects of dispersal. In contrast, differences among host lineages primarily strengthened ecological selection associated with host identity [10,11].

The plant microbiome has been proposed as a "second genome" that complements the host genome, and vertically transmitted seed microbes constitute a potentially heritable component of this microbial genome [1,6,20]. Having established strong host filtering in seed microbiome assembly, we asked whether this microbial inheritance also encodes host identity. Indeed, host genus could be reliably inferred from seed microbiome composition, with adjusted Rand indices (ARIs) of 0.421–0.663 across

bacterial V4, V5–V7, and fungal ITS1 datasets (Fig. 3D–F). This predictability is consistent with experimental evidence that host identity can strongly influence microbiome assembly across broad environmental gradients [43], although the strength of host filtering varies among plant species [44]. Notably, host identity could be recovered without prior knowledge of host labels during clustering, indicating that seed microbiomes contain robust information about their hosts. Importantly, this host-associated signal was concentrated in fine-scale microbial variation: collapsing ASVs to the genus level reduced ARI to 0.189 for 16S and 0.212 for ITS, approximately one-third of the original values (Fig. S6). This pattern supports evidence that coarse taxonomic classification can obscure strain-level information associated with vertical transmission [45], consistent with the very small seed core and genotype-specific microbiome reported in tomato [10]. Thus, the ability of seed microbiomes to encode host identity may depend strongly on fine-scale microbial variation, suggesting that closely related microbial lineages can represent distinct ecological entities whose host-associated specialization is obscured by coarse taxonomic aggregation. Together, these findings support the view that the seed microbiome acts as a microbial extension of the host phenotype [46].

Generalists and specialists in the seed microbiome showed distinct patterns of host association rather than occurring randomly across hosts. Generalists were mainly concentrated in a small number of widespread, aerobic, and biofilm-forming lineages. In contrast, specialists were distributed across a much broader range of microbial lineages and were more frequently associated with anaerobic taxa. These differences were particularly pronounced in bacteria but were weaker in fungi (Fig. 3, Fig. S8). These patterns echo multidimensional specialization and generalization in soil prokaryotes, where taxa tend to follow distinct ecological trajectories rather than occupy niches randomly [27,47]. Heterospecific transmission experiments suggest that taxa capable of resisting host filtering are favored across heterogeneous hosts [48], while global metagenomic analyses indicate that generalists persist across diverse ecological constraints through greater adaptability, whereas specialists remain more tightly

391 constrained by environmental selection [49]. Conversely, strong host-mediated selection
392 for attachment and resource acquisition may favor specialist lineages [50,51], consistent
393 with host filtering and identity as major determinants of microbial assembly in other
394 host-associated systems [17,52]. Host breadth therefore reflects contrasting ecological
395 constraints on microbial persistence.

396 This overall pattern was also evident at the lineage level, where several
397 phylogenetically related taxa were disproportionately represented among generalists
398 and provided potential ecological bases for their broad host distributions. The strong
399 enrichment of *Pseudomonas*, *Pantoea*, *Massilia*, *Paenibacillus* and other
400 Gammaproteobacteria among generalists suggests that broad host distribution is partly
401 rooted in lineage-specific ecological versatility (Fig. S8A, B). *Pseudomonas* is
402 characterized by diverse and adaptable genomes that support metabolic and regulatory
403 flexibility across fluctuating environments [53], while *Pantoea* shows similarly broad
404 ecological distributions and a wide range of plant-associated functions [54]. *Massilia* can
405 efficiently colonize plant roots and deploy multiple pathways to tolerate and modify
406 host-derived chemical constraints [55], further illustrating how flexible resource use and
407 host interaction can promote persistence across heterogeneous environments. These
408 lineage-level characteristics closely matched the functional signature of our generalists,
409 which were predominantly aerobic, biofilm-forming, motile and stress-tolerant (Fig. 5,
410 Fig. S10). Such traits are relevant to repeated transitions among hosts, as biofilm
411 formation can promote stable plant colonization [56], while extracellular matrix
412 components in seeds may enhance bacterial persistence and resource acquisition [57].
413 Their larger genomes and expanded repertoires of transporters and regulators further
414 suggest greater capacity to sense and exploit heterogeneous host environments,
415 consistent with links between genomic breadth and habitat occupancy [33,58] and between
416 genome size and generalist strategies [34,47]. By contrast, specialists were
417 phylogenetically dispersed and enriched in anaerobic and maintenance-related
418 functions, indicating adaptation to narrower ecological contexts rather than uniformly
419 reduced functional capacity. Thus, generalism appears to reflect the convergence of

lineage-specific ecological versatility and a coordinated suite of persistence and resource-acquisition traits, whereas specialist diversity may arise through a wider range of finely tuned ecological solutions.

Host breadth and abundance capture different dimensions of ecological success. Although generalists tended to be more abundant overall, specialists can also achieve high abundance in suitable hosts (Fig. S9). The high cumulative abundance of generalists may instead reflect repeated recruitment across many hosts under predominantly neutral assembly processes [51,59], rather than superior competitive ability within any individual host. Specialists can likewise become locally abundant when conditions match their narrower requirements, and ecological specificity can be closely associated with abundance within suitable habitats [60]. Moreover, low-abundance taxa can exert disproportionate effects on ecosystem functioning [61]. Thus, broad host range reflects repeated recruitment across hosts, not inherently greater ecological importance.

Specialization was most evident among microbial groups whose persistence depended on specific physiological conditions or host-mediated interactions. Mycorrhizal fungi concentrated in a limited set of reservoir hosts (Fig. S12) exemplify specialization driven by symbiotic compatibility: ectomycorrhizal fungi often show strong host preferences, while host–symbiont coevolution and resource exchange can constrain compatible associations [62]. Anaerobic bacteria were concentrated in *Gossypium*, *Achnatherum*, and *Brassica* (Fig. S13), consistent with constraints imposed by oxygen availability and redox conditions that limit anaerobic metabolism [63]. Minimal-genome lineages face another constraint: genome streamlining reduces biosynthetic capacity, increasing dependence on host-derived metabolites or co-resident microbes [64,65]. Together, these patterns suggest that specialization emerges when microbial persistence requires a narrow set of physiological conditions or host-mediated interactions. Nitrogen fixation, however, provided a notable exception, with this function retained across relatively broad host ranges. Overall, 31.8% of nitrogen-fixing ASVs fell within the generalist–intermediate breadth range, more than twice the proportion observed among anaerobic bacteria (15.2%) and more than 20-fold that

among mycorrhizal fungi (1.3%) (Fig. S11), and broad-host-range diazotrophs included *Azospirillum* and *Herbaspirillum* (Fig. S14). This broader distribution may reflect the multiple ecological modes through which nitrogen fixation is realized: symbiotic fixation depends on host–microbe signaling and nodule development and can involve strong host specificity [66], whereas associative, endophytic, and free-living diazotrophy does not require an obligate host-specific interaction; aerobic endophytes can fix nitrogen in non-nodulating plants [67], and leaf-associated diazotrophs occur across diverse hosts [68]. Thus, the breadth of a microbial function depends not only on its ecological importance, but also on the mode through which it is realized.

Several limitations should be considered. Our synthesis combines studies differing in sequencing regions, sample processing, and ecological context, although consistent patterns across three amplicon regions and variance partitioning reduced the influence of methodological heterogeneity. Functional traits and gene repertoires were also inferred rather than directly measured, and host breadth was estimated from observed occurrence rather than experimental fitness. Future studies integrating genome-resolved metagenomics, culture-based assays, reciprocal inoculations, and longitudinal sampling could test whether traits associated with generalism directly enhance colonization across hosts and how host filtering changes during seedling development [69,70].

Overall, our synthesis shows that seed microbiomes are structured by reproducible host filtering rather than random environmental recruitment. Generalists tend to combine broad resource-use and persistence traits, whereas specialists occupy narrower physiological or host-dependent niches. Host identity and microbial life-history strategy therefore represent complementary axes shaping the assembly and functional organization of seed microbiomes, providing a framework for understanding microbial persistence and specialization across plant hosts.

Materials and Methods

Literature search and dataset assembly

To compile a comprehensive seed microbiome dataset, we systematically searched Google Scholar and Web of Science to July 2026 using the query ("seed" OR "seeds" OR "seed-borne" OR "seed-associated") AND ("microbiome" OR "microbiota" OR "endophyte" OR "microbial community" OR "bacteriome" OR "mycobiome"). Studies were included if they were peer-reviewed, surface-sterilised seeds before microbiome profiling, used 16S rRNA or ITS amplicon sequencing of surface-sterilised seeds or aseptically germinated seedlings, deposited raw data in SRA/ENA, and reported host species. Samples were classified as type S (directly extracted after sterilisation) or type P (aseptically germinated before extraction), yielding 2,722 S-type and 1,385 P-type 16S samples and 2,076 S-type and 291 P-type ITS samples after quality control. Overall, 58 studies met these criteria (S1–S58; Table S1), including 51 16S and 26 ITS studies, spanning 21 countries for 16S and 30 for ITS.

To broaden host phylogenetic and geographic coverage, we additionally incorporated the dataset of Simonin et al. (2022; S59), comprising 1,371 16S samples from 62 host species across 29 countries and 907 ITS samples from 52 host species across 26 countries [8]. Its prebuilt OTU/ASV tables covering four amplicon regions (V4, V5–V6, ITS1 and ITS2) were integrated during subsequent region-specific merging. In total, 59 studies were included.

Seed microbial extraction and sequencing (in-house)

To expand the seed microbiome dataset beyond published studies, we additionally generated an in-house, as yet unpublished amplicon dataset (S60) with a consistent quality-control standard. 52 species (52 for 16S; 51 for ITS) were sequentially surface-sterilised by three washes in sterile distilled water followed by a 5% sodium hypochlorite immersion for 5 min, three washes in sterile distilled water followed by a 75% ethanol immersion for 2 min, and five washes in sterile distilled water. Surface-

sterilised seeds were placed on sterile gauze moistened with sterile distilled water in Petri dishes and incubated in the dark. Germinating seeds were harvested for DNA extraction, tissue was ground in liquid nitrogen, and total DNA was extracted using a Magnetic Soil and Stool DNA Kit (DP336-02; TianGen, Beijing, China). Bacterial communities were profiled by amplifying the 16S rRNA V5 – V7 region (799F: AACMGGATTAGATACCCKG; 1193R: ACGTCATCCCCACCTTCC; 258 samples), and fungal communities were profiled by amplifying the ITS1 region (ITS1: CTTGGTCATTTAGAGGAAGTAA; ITS2: GCTGCGTTCTTCATCGATGC; 235 samples); amplicon libraries were sequenced on the Illumina paired-end platform. All S60 samples correspond to extraction type P (sterile germination followed by extraction from germinated tissue).

Amplicon data processing and cross-region integration

Raw amplicon data from three sources were processed and integrated: 58 literature-derived studies, the seed microbiome dataset of Simonin et al. (2022) (S59) [8], and our in-house sequencing data (S60). For the first and third sources, reads were assigned to five primary amplicon regions (V3–V4, V4, V5–V7, ITS1, and ITS2); V5–V6 data were additionally incorporated from the prebuilt tables of S59. Reads were quality-filtered and primer-trimmed with fastp (v1.0.0) [71] and Cutadapt (v2.7) [72], followed by ASV inference, chimera removal, and taxonomy assignment in VSEARCH (v2.30.0) [73] using SILVA v138.2 [74].

To integrate ASVs generated from different primer regions, representative sequences were mapped to SILVA (16S; v138.2) or UNITE (ITS; v25.2) [75] reference sequences at ≥97% identity. ASVs mapping to the same reference sequence were aggregated under a common TAXID, generating sample × TAXID abundance matrices while retaining unmapped ASVs. Mapped sequences inherited taxonomy from the reference database, whereas unmapped sequences were independently classified with VSEARCH. Overall, 65.6% of 16S and 58.2% of ITS ASVs lacked a ≥97% reference match, but represented only 32.6% and 32.2% of reads, respectively. The prebuilt ASV

tables of S59 were similarly re-mapped to the same reference databases and converted to TAXID-based matrices for integration.

During literature curation, only intra-seed microbial samples were retained, including mature seeds and sterilely germinated seedlings derived from surface-sterilised seeds. Rhizosphere, phyllosphere, seed-surface/seed-coat samples, and samples lacking clear host identities were excluded. Samples with <1,000 reads, non-target sequences, chloroplast/mitochondrial 16S reads, or fewer than two detected genera were removed, and remaining data were normalised by cumulative sum scaling (CSS). The final dataset comprised 4,107 bacterial and 2,367 fungal samples. To minimise biases associated with differences among primer regions, primary analyses were conducted separately by amplicon region; cross-region integration was used only for genus-level robustness analyses.

Host-level presence/absence matrix and α diversity

Abundance information was retained throughout the analysis, with CSS-normalised relative-abundance tables used for all α-diversity, β-diversity, and community-assembly analyses. Host breadth was quantified separately from species-level presence/absence profiles derived from the TAXID abundance matrices. Within each host species, taxon prevalence was calculated across biological replicates, and a species-specific richness threshold (T) was defined as the mean or median sample richness: for species with ≥10 replicates, the mean was used when richness was normally distributed (Anderson–Darling, $p > 0.05$) and the median otherwise; for species with <10 replicates, the mean was used. Taxa were ranked by prevalence, and the top T taxa were retained as core members of each host species. The resulting species × TAXID matrix was used to calculate cross-species prevalence as the number of host species harbouring each taxon (16S: 11,033 taxa; ITS: 2,144).

Four α-diversity indices (Shannon, Simpson, observed ASV richness, and Pielou's evenness) were calculated from region-specific CSS-normalised tables. Differences among host genera, families, and orders were tested using Kruskal–Wallis tests within

each amplicon region.

Host phylogeny and phylogenetic signal

To test whether host evolutionary relatedness shapes seed microbial α diversity, we constructed a time-calibrated host plant phylogeny by submitting all host species names to the TimeTree5 [76] platform and downloading the resulting tree (176 host species; a genus-level tree of 123 host genera was used for genus-level analyses). The species tree was pruned to the host species represented in each region (88 / 74 / 103 tips for V4 / V5–V7 / ITS1). To quantify the degree to which α diversity is evolutionarily conserved, we computed Blomberg`s K and Pagel`s λ ("phytools::phylosig", test = TRUE; λ by maximum likelihood, K by randomisation with 999 permutations)[77,78] for each region × α-index combination. To account for the sampling error arising from estimating species means from unequal sample sizes, we resampled samples within each species with replacement (B = 500, seeded for reproducibility), re-estimated λ at each iteration, and reported the 2.5th and 97.5th percentiles as a bootstrap 95% confidence interval.

β diversity and host genetic distance

To assess phylosymbiosis, sample-level Bray–Curtis dissimilarities were calculated for each amplicon region and averaged across all sample pairs between host species. Host genetic distance was obtained from the TimeTree5 phylogeny, and the association between host relatedness and microbiome dissimilarity was tested using Mantel tests and multiple regression on distance matrices (MRM; 999 permutations).

To quantify the contributions of host identity and geographic origin, sample-level Bray–Curtis distances were analysed using marginal PERMANOVA and variance partitioning, with host genus, source country, and extraction method as predictors. Because host identity and geography are partially confounded in the observational dataset, these estimates were complemented by controlled contrasts described below

(see "Community assembly processes"). As a robustness check, ASVs were additionally aggregated to microbial genus and all regions were merged before repeating the analyses, with amplicon region included as an additional factor.

Community assembly processes

To quantify the relative contributions of deterministic and stochastic processes to seed microbiome assembly, we applied the iCAMP framework (v1.5.12) [79]. For sample pairs within each region, βNTI and Bray–Curtis-based Raup–Crick (RCbray) were calculated within genus × extraction and country × extraction bins and used to classify five assembly processes: heterogeneous selection ($\beta NTI > 2$), homogeneous selection ($\beta NTI < -2$), dispersal limitation ($|\beta NTI| \leq 2$ and $RCbray > 0.95$), homogenising dispersal ($|\beta NTI| \leq 2$ and $RCbray < -0.95$), and drift (otherwise). Process proportions were calculated within each bin to account for unequal sample sizes.

We further used three controlled contrasts to separate host and geographic effects while holding technical covariates constant. First, host pairs were grouped by their most recent common taxonomic level (same species to cross-phylum) to test whether deterministic selection increased with host phylogenetic distance, using contingency-based tests and within-country permutations. Second, among same-genus, same-extraction pairs, same- versus cross-country comparisons isolated geographic effects on assembly processes. Third, an S60 subset of in-house samples from the same batch, extraction, and country (China) was used to isolate the effect of host taxonomy on assembly.

Host identity prediction and ARI clustering

To test whether seed microbiome composition alone can recover host identity, we performed unconstrained hierarchical clustering ("stats::hclust") on the sample-level Bray-Curtis dissimilarity of each region over a full grid of eight linkage methods (ward.D, ward.D2, single, complete, average, mcquitty, median, centroid) and k = 2–400, without a dissimilarity threshold (T = 0). At each grid point we recomputed a set

of external validation metrics against host labels at four taxonomic levels (order, family, genus, species), including the adjusted Rand index (ARI), normalised mutual information, weighted purity, homogeneity, completeness, V-measure, and host splitting; the silhouette coefficient (an internal metric that uses no host information) was recorded for comparison. The ARI corrects for the chance agreement expected of a random partition, so ARI values well above 0 indicate that clustering is genuinely driven by host identity. We selected two representative partitions per region: the best-ARI partition (highest ARI against host genus) and the min-K partition (smallest k attaining ARI $\geqslant$ 0.20). To test whether host-identity prediction depends on microbial taxonomic resolution, the genus-merged whole tables (16S n = 4,082; ITS n = 2,352) were re-analysed under the identical grid and validation protocol.

Identification of ecological generalists and specialists

To rigorously classify microbial taxa as ecological generalists or specialists, we applied a two-stage double screening based on the host species-level presence/absence matrix. In the first stage, to determine a breadth threshold, we constructed a within-column permutation null model that preserves each host species" representative richness (T): for each column, T taxa were randomly drawn over 1,000 iterations to generate the null distribution of cross-species prevalence, and the smallest k at which the observed count first exceeded the null mean was taken as the generalist threshold (*n_hosts* = 5 for both 16S and ITS).

In the second stage, to exclude "pseudo-generalists" that are broad yet phylogenetically clustered, we computed the host phylogenetic dispersion of each taxon: for each prevalence k (k $\geqslant$ 2), k host species were randomly drawn 1,000 times and the mean pairwise host phylogenetic distance (MPD) was computed to build a per-k null distribution; the standardised z-score was then calculated and tested one-sided for dispersion and clustering, each with Benjamini – Hochberg FDR correction. Classification rules were: generalist = *n_hosts* $\geqslant$ 5 and not significantly clustered

(q_clus ⩾ 0.05); specialist = *n_hosts* = 1; intermediate = the remainder. This double screening removed 32 (16S) and 41 (ITS) high-breadth but significantly clustered pseudo-generalists, yielding final generalist counts of 1,103 (16S) and 235 (ITS) and specialist counts of 5,830 (16S) and 1,232 (ITS). For the clustered taxa retained by this screening, the representative host genus (the genus most frequently hosting each taxon) was tabulated (Table S4).

Phylogenetic trees and composition of generalists and specialists

To place generalist and specialist taxa in an evolutionary framework, we constructed separate maximum-likelihood phylogenetic trees for bacterial (16S) generalist (1,103 sequences) and specialist (5,830 sequences) taxa. Generalist sequences were aligned with MUSCLE (v5.3) [80], trimmed with trimAl (v1.5; "-automated1") [81], and the tree was inferred with IQ-TREE (v3.0.175) [82] under ModelFinder model selection with 1,000 ultrafast bootstrap replicates. Specialist sequences were aligned with mafft ("--auto") [83], trimmed with trimAl, and the tree was inferred with FastTree (v2.2.0; "-nt -gtr -gamma") [84]. Trees were visualised in iTOL (v7) [85], with annotation files generated using the itol.toolkit R package (v1.1.2) [86]: the generalist tree was overlaid with phylum/class colour strips, BugBase-predicted trait rings (aerobic type, biofilm formation, Gram type), and an outer *n_hosts* bar; the specialist tree was annotated analogously.

To characterise compositional differences between generalists and specialists, we compared taxon representation across major taxonomic ranks using Fisher's exact tests, with BH-FDR correction; taxa occurring exclusively in either group were reported separately. At the genus level, ASVs were aggregated and genera ranked by host breadth (*n_hosts* and *n_hostg*), with the 25 most broadly distributed genera reported together with their mean relative abundance. Finally, within the major ASV-rich phyla (16S) and classes (ITS), we tested the relationship between host breadth and non-zero relative abundance using linear regression and one-sided Spearman correlations.

Functional traits and genes

All functional traits and gene content were inferred from marker-gene data by in silico prediction and are therefore referred to as predicted traits. Predicted functional traits were compiled as ASV × trait tables from four resources: BugBase (v0.1.0; 16S; Greengenes v13.8) [87,88] , covering aerobic type, biofilm formation, Gram type, stress tolerance, mobile elements and potential pathogenicity; FAPROTAX (v1.2.12; 16S) [89], covering ecological functions including human association, plant pathogenicity, nitrate reduction and nitrogen respiration; metaTraits (v0.1.14; 16S; GTDB-based) [90,91], providing continuous traits (temperature, pH, salinity, genome size, gene count, GC content and coding density) and binary traits (carbon-source utilisation, enzyme activity, motility, gas production, Gram staining and others); and FUNGuild (v1.1; ITS) [92] , providing seven fungal guild traits (Pathotroph, Pathogen, Endophyte, Saprotroph, Saprotrophs, Symbiotroph and Mycorrhizal). Trait columns were classified as binary (TRUE/FALSE) or numeric. Binary traits were analysed among annotated ASVs (NAs excluded) using binomial GLMs against log2(*n_hosts* + 1), with Wald tests and BH-FDR correction; traits with <100 annotated ASVs or no variation were excluded. Effect sizes were summarized as rank-biserial correlations and odds ratios (ORs) per doubling of host breadth with 95% confidence intervals. Numeric traits were tested using Spearman correlations with *n_hosts* and BH-FDR correction. Overall, 115 binary traits (108 bacterial, 16S; 7 fungal, ITS) and 15 continuous bacterial traits were screened, with *n_hosts* capped at 20 to reduce long-tail influence.

To characterise gene-level functional differentiation, KO presence/absence was predicted for all 16S ASVs using PICRUSt2 v2.6.2 (--max_nsti 0.15) [93]. KOs present in all or no ASVs were excluded, leaving 9,580 KOs for association testing. KO occurrence was modelled against log2(*n_hosts* + 1) using the same logistic framework as for binary traits, with likelihood-ratio tests and BH-FDR correction. Wald odds ratios (ORs) with 95% confidence intervals were used to estimate effect sizes. Significant KOs (FDR < 0.05) were classified as generalist-enriched (log2 OR ≥ 0.5) or specialist-enriched (log2 OR ≤ −0.5), with weaker effects left unassigned. KO families were annotated to KEGG level-2 categories [94] and five-level plant-growth-promoting gene

(PGPG) functions using annotations derived from the PGPg_finder database [95,96].

Special classification groups

To examine the distribution of taxa with distinctive ecological or functional attributes, we constructed four special classification groups from the predicted functional annotations: (i) Mycorrhizal fungi = FUNGuild "Mycorrhizal == TRUE" (ITS, 227 ASVs); (ii) Anaerobic bacteria = BugBase "Anaerobic == TRUE" (16S, 3,479 ASVs); (iii) Nitrogen-fixing bacteria = FAPROTAX "nitrogen_fixation == TRUE" (16S, 220 ASVs); and (iv) Small-genome bacteria = the smallest 1% of non-NA metaTraits genome sizes (16S, 108 ASVs). For each group we quantified ASV counts, sample-level relative-abundance distributions, order-level taxonomic composition and generalist/specialist composition, and host loading (the number of group member genera detected per host genus, and its correlation with group relative abundance). To visualise group distribution at genus resolution, group ASVs were annotated to the finest resolvable taxonomic level and aggregated into a label × host genus mean relative-abundance matrix (labels detected in ⩾ 3 host genera), which was log10-transformed, row-scaled (z-score), and double-clustered (ward.D2) with per-genus × host-order Kruskal-Wallis tests (BH-FDR corrected).

Statistical analysis

All data manipulation and statistical analyses were performed in R (v4.5.2; R Foundation for Statistical Computing, Vienna, Austria). Community analyses (PERMANOVA/adonis2, Mantel, variance partitioning, diversity) used the vegan package; phylogenetic signal (Pagel`s λ, Blomberg`s K) used phytools/ape; MRM used ecodist; hierarchical clustering and ARI used stats and the cluster metrics described above; βNTI/RCbray were computed with the iCAMP workflow on the server; heatmaps used ComplexHeatmap/pheatmap; and visualisation used ggplot2, ggtree, and patchwork. The default significance threshold was $\alpha = 0.05$; multiple

comparisons were corrected with the Benjamini–Hochberg FDR. Permutation-based tests (Mantel, MRM, adonis2, X^2 with Monte Carlo simulation, within-bin permutation, MPD null model, within-column null model) were run with 999 or 1,000 iterations as stated, and the λ bootstrap with B = 500. Effect sizes were reported as appropriate (ARI, Mantel r, MRM R^2, marginal R^2, Cramer`s V, Spearman ρ, logistic odds ratios, rank-biserial ρ, Pagel·s λ). Sample sizes after quality control were 4,107 (16S) and 2,367 (ITS). p values below 0.001 are reported as $p < 0.001$.

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Author contributions:

N.L. conceived the study. Z.W. performed the meta-analysis, including data analysis, visualization, and drafting of the manuscript. W.A., X.L., Y.X., and P.H. contributed to discussions on data analysis. T.M., and N.L. reviewed and revised the manuscript. All authors reviewed and edited the final manuscript.

Competing interests:

Authors declare that they have no competing interests.

Data, code, and materials availability:

All data needed to evaluate the conclusions in the paper are present in the paper and/or the supplementary materials. Original sequence data used in the meta-analysis can be accessed using the accession numbers provided in the *metadata_16s.txt* and *metadata_its.txt* files in the GitHub repository or the project accession numbers listed in Supplementary Table 1. Newly generated bacterial and fungal amplicon sequencing data have been deposited in the National Genomics Data Center (NGDC) BioProject database under accession number CRA044477 OR PRJCA066385 (Temporary links for sequence access: https://ngdc.cncb.ac.cn/gsa/s/KoWEE8c8). All code used in this study is publicly available at https://github.com/WTQMBGMB/Large-scale-meta-analysis-of-seed-microorganisms/tree/main

Figures

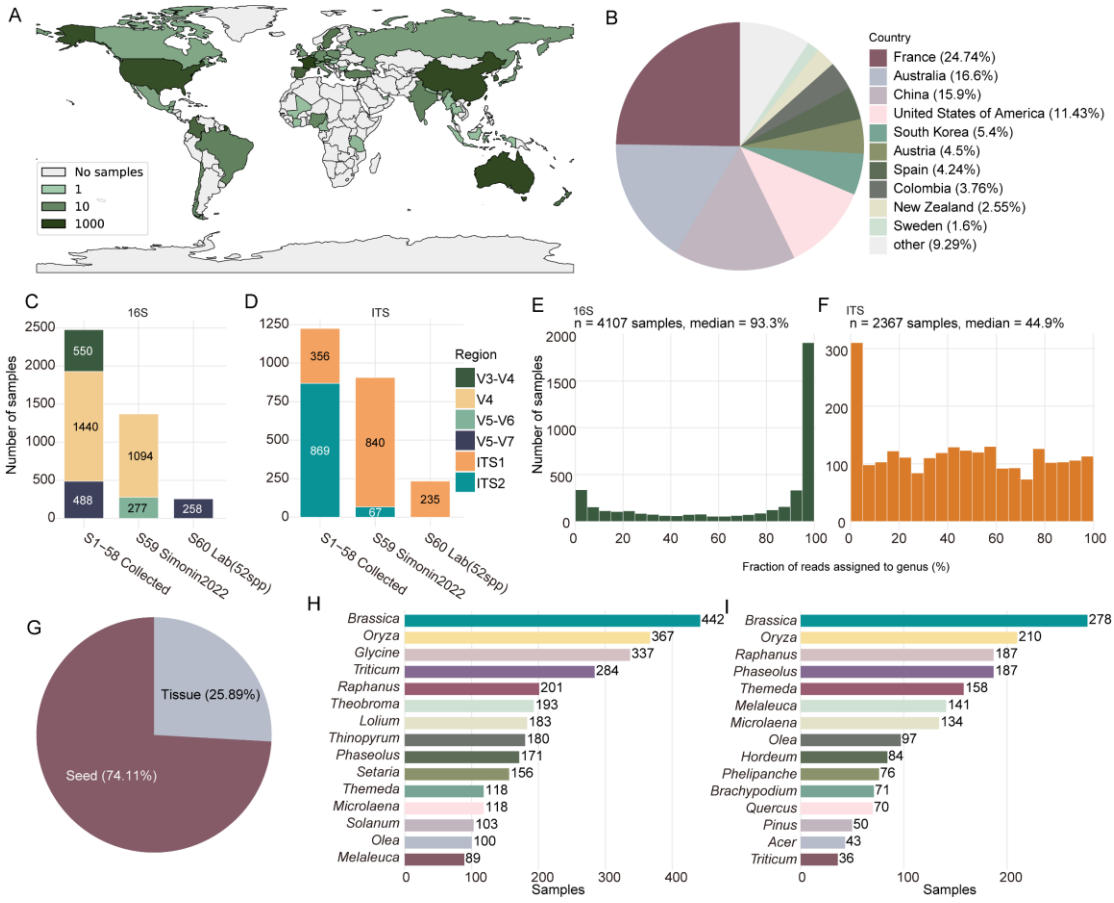
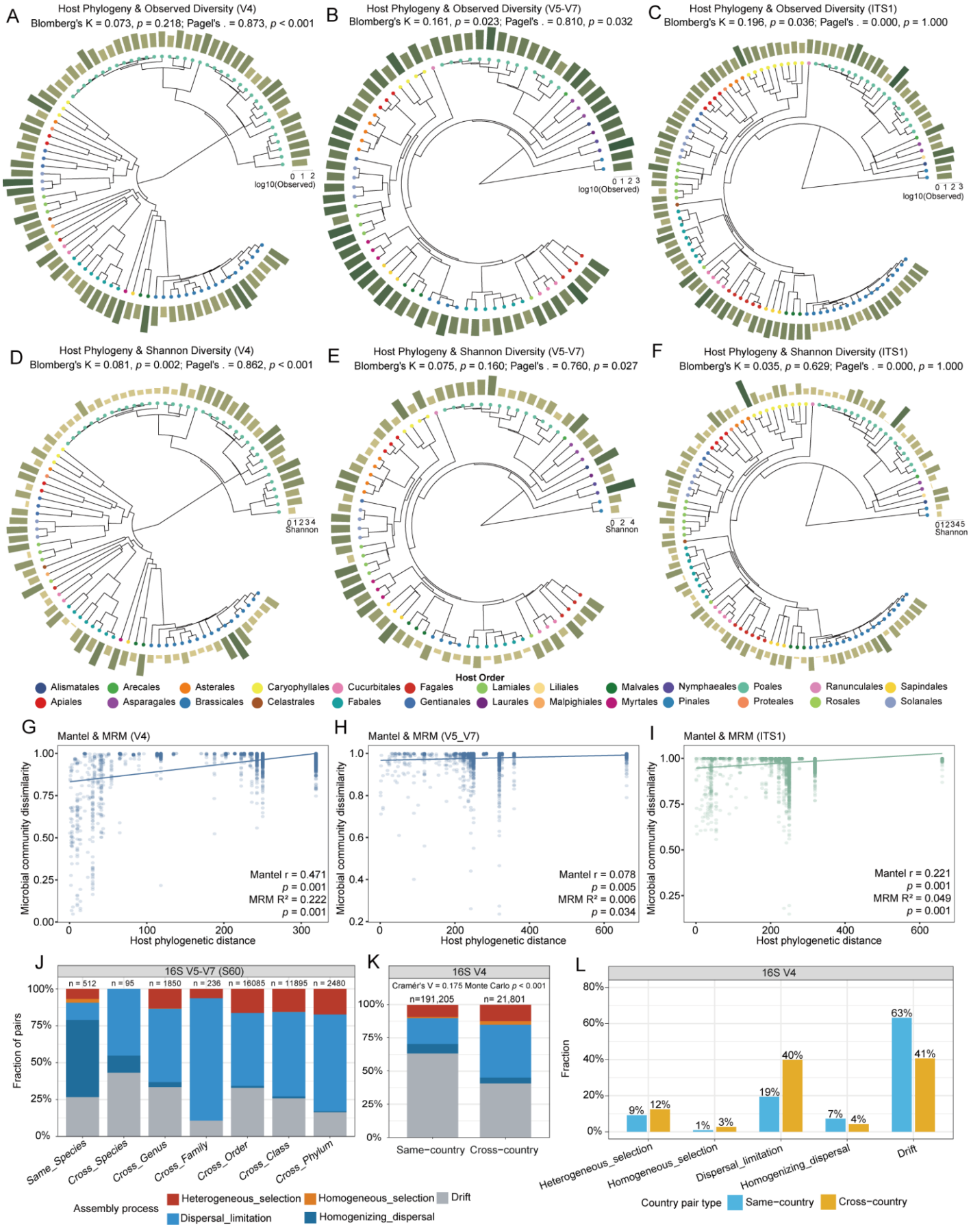
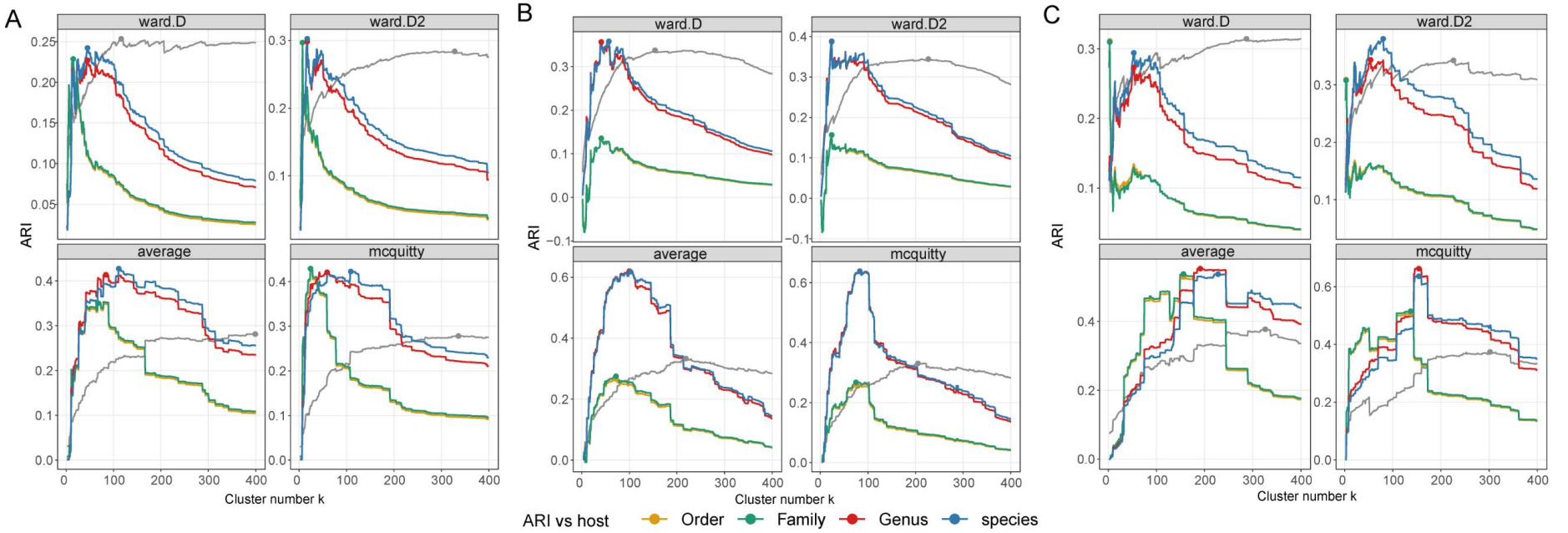


Figure 1 Global overview of the seed microbiome dataset. (A) Geographic distribution of the 6,440 amplicon samples with an assignable country, coloured by the log10-transformed number of samples per country (34 samples without a country assignment are not shown). (B) Country composition, showing the ten most represented countries (France, Australia, China and the United States together represent 68.7% of samples). (C, D) Sample counts per data source (S1–S58, literature; S59, Simonin et al. 2022; S60, this study) stacked by amplicon region for the 16S (C) and ITS (D) datasets. (E, F) Distribution of the proportion of reads assigned to genus level per sample in the 16S (E; n = 4,107; median = 93.3%) and ITS (F; n = 2,367; median = 44.9%) datasets. (G) Sample types (seed, 74.1%; tissue, 25.9%). (H, I) The 15 most sampled host genera in the 16S (H) and ITS (I) datasets.


A
Host Phylogeny & Observed Diversity (V4)
Blomberg's K = 0.073, p = 0.218; Pagel's . = 0.873, p < 0.001
log10(Observed)
B
Host Phylogeny & Observed Diversity (V5-V7)
Blomberg's K = 0.161, p = 0.023; Pagel's . = 0.810, p = 0.032
log10(Observed)
C
Host Phylogeny & Observed Diversity (ITS1)
Blomberg's K = 0.196, p = 0.036; Pagel's . = 0.000, p = 1.000
log10(Observed)
D
Host Phylogeny & Shannon Diversity (V4)
Blomberg's K = 0.081, p = 0.002; Pagel's . = 0.862, p < 0.001
Shannon
E
Host Phylogeny & Shannon Diversity (V5-V7)
Blomberg's K = 0.075, p = 0.160; Pagel's . = 0.760, p = 0.027
Shannon
F
Host Phylogeny & Shannon Diversity (ITS1)
Blomberg's K = 0.035, p = 0.629; Pagel's . = 0.000, p = 1.000
Shannon
Host Order
Alismatales
Apiales
Arecales
Asparagales
Asterales
Brassicales
Caryophyllales
Celastrales
Cucurbitales
Fabales
Fagales
Gentianales
Lamiales
Laurales
Liliales
Malpighiales
Malvales
Myrtales
Nymphaeales
Pinales
Poales
Proteales
Ranunculales
Rosales
Sapindales
Solanales
G
Mantel & MRM (V4)
Microbial community dissimilarity
Host phylogenetic distance
Mantel r = 0.471
p = 0.001
MRM R² = 0.222
p = 0.001
H
Mantel & MRM (V5_V7)
Microbial community dissimilarity
Host phylogenetic distance
Mantel r = 0.078
p = 0.005
MRM R² = 0.006
p = 0.034
I
Mantel & MRM (ITS1)
Microbial community dissimilarity
Host phylogenetic distance
Mantel r = 0.221
p = 0.001
MRM R² = 0.049
p = 0.001
J
16S V5-V7 (S60)
n = 512
n = 95
n = 1850
n = 236
n = 16085
n = 11895
n = 2480
Fraction of pairs
Same_Species
Cross_Species
Cross_Genus
Cross_Family
Cross_Order
Cross_Class
Cross_Phylum
K
16S V4
Cramér's V = 0.175 Monte Carlo p < 0.001
n=191,205
n= 21,801
Same−country
Cross−country
L
16S V4
Fraction
9%
12%
1%
3%
19%
40%
7%
4%
63%
41%
Heterogeneous_selection
Homogeneous_selection
Dispersal_limitation
Homogenizing_dispersal
Drift
Country pair type
Same−country
Cross−country
Assembly process
Heterogeneous_selection
Homogeneous_selection
Drift
Dispersal_limitation
Homogenizing_dispersal

1081

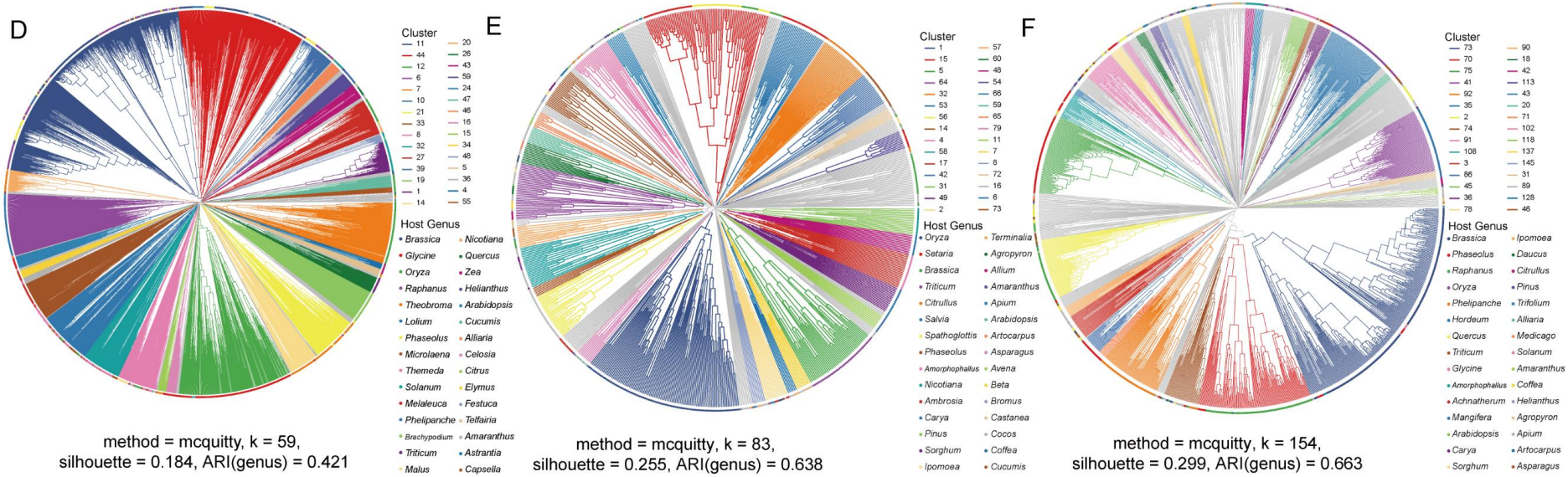
Figure 2 Host phylogeny and identity shape seed microbial diversity, composition and assembly. (A–C) Host phylogenies for the V4 (A), V5–V7 (B) and ITS1 (C) datasets, with tips coloured by host order and outer bars showing species-mean observed ASV richness (log10; 88, 74 and 103 species, respectively). Pagel's λ is indicated. (D–F) As in A–C, for Shannon diversity, with Pagel's λ indicated. (G–I) Host phylogenetic distance versus species-level mean Bray–Curtis dissimilarity for V4 (G), V5–V7 (H) and ITS1 (I), with Mantel and multiple regression on distance matrices (MRM) statistics indicated. (J) Relative contributions of five inferred community-assembly processes across seven levels of host-lineage divergence in the laboratory-controlled S60 V5–V7 dataset. (K) Assembly-process composition for V4 sample pairs within host-genus pools, comparing same-country (n = 191,205 pairs) and cross-country (n = 21,801 pairs) comparisons. (L) Relative proportions of individual assembly processes for the comparisons shown in K. V5–V7 cross-country comparisons are not shown because only 132 pairs were available.



Clustering tree — V4 region (n = 2450 samples, 59 clusters)

Clustering tree — V5-V7 region (n = 745 samples, 83 clusters)

Clustering tree — ITS1 region (n = 1413 samples, 154 clusters)



method = mcquitty, k = 59,
silhouette = 0.184, ARI(genus) = 0.421

method = mcquitty, k = 83,
silhouette = 0.255, ARI(genus) = 0.638

method = mcquitty, k = 154,
silhouette = 0.299, ARI(genus) = 0.663

Figure 3 Hierarchical clustering of seed microbial community dissimilarity recovers host taxonomic structure. (A–C) Agreement between hierarchical clustering of Bray–Curtis dissimilarity and host taxonomy across cluster numbers k (2 ~ 400) for the V4 (A; n = 2,450), V5–V7 (B; n = 745) and ITS1 (C; n = 1,413) datasets (four of the eight linkage methods tested; silhouette width in grey and adjusted Rand index, ARI, against host Order/Family/Genus/species labels in colour). ARI is corrected for chance (0 = agreement expected by chance, 1 = perfect match). The best ARI against host genus is far above 0 in all datasets , 0.421 (A; mcquitty, k = 59), 0.638 (B; k = 83) and 0.663 (C; k = 154) , at k of the same order as the number of host genera (58, 64 and 85), whereas the silhouette-optimal partitions recover host genus only weakly (ARI = 0.117, 0.202 and 0.435). (D–F) Dendrograms of the best-ARI partitions (mcquitty; k = 59, 83 and 154), branches coloured by cluster and tips by host genus. The coherent colour blocks show that clusters largely follow host genus lines, supporting the seed microbiome as a host phenotypic trait.

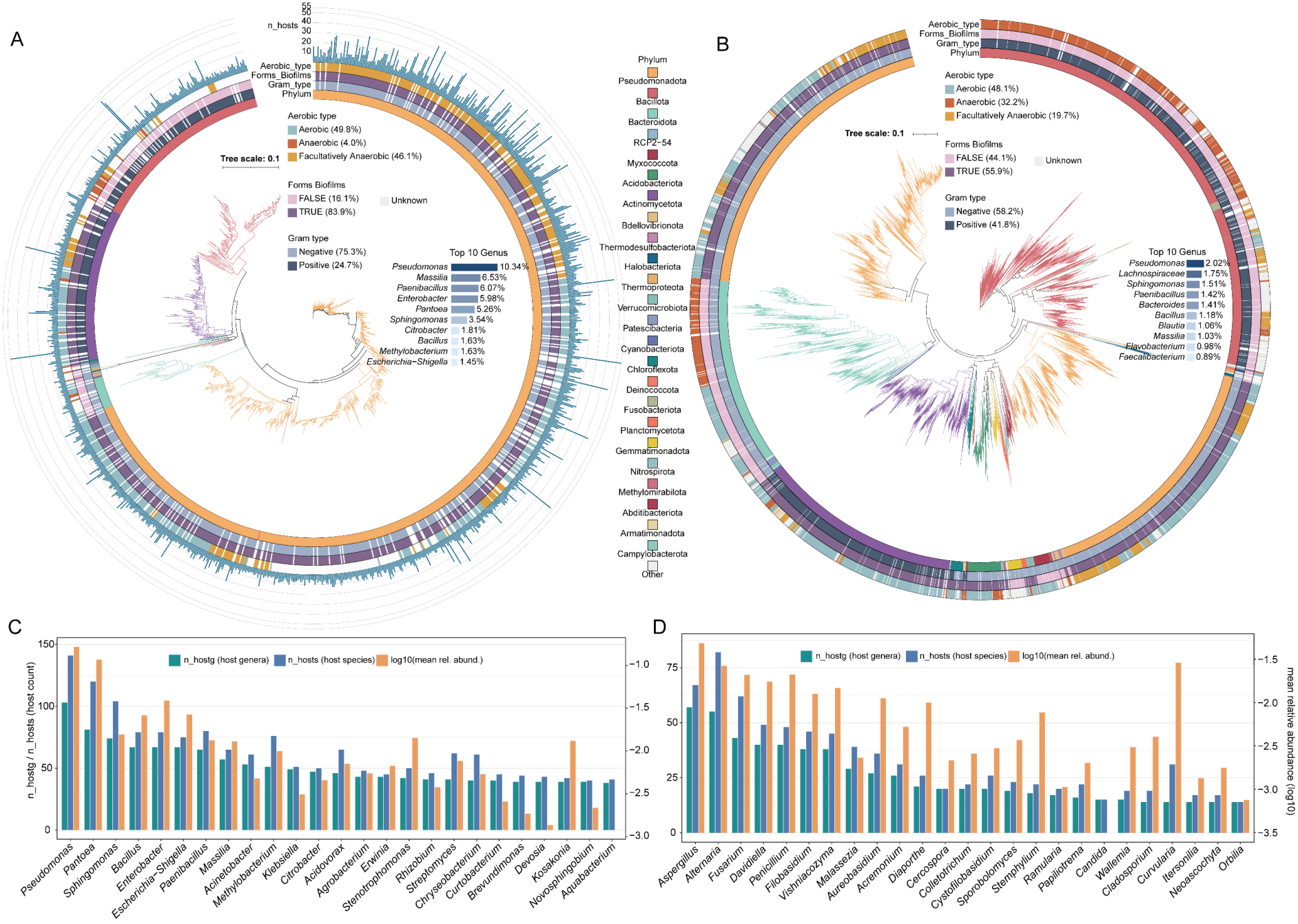

A
n_hosts
55
50
40
30
20
10
Aerobic_type
Forms_Biofilms
Gram_type
Phylum
Aerobic type
Aerobic (49.8%)
Anaerobic (4.0%)
Facultatively Anaerobic (46.1%)
Tree scale: 0.1
Forms Biofilms
FALSE (16.1%)
TRUE (83.9%)
Unknown
Gram type
Negative (75.3%)
Positive (24.7%)
Top 10 Genus
Pseudomonas 10.34%
Massilia 6.53%
Paenibacillus 6.07%
Enterobacter 5.98%
Pantoea 5.26%
Sphingomonas 3.54%
Citrobacter 1.81%
Bacillus 1.63%
Methylobacterium 1.63%
Escherichia−Shigella 1.45%
Phylum
Pseudomonadota
Bacillota
Bacteroidota
RCP2−54
Myxococcota
Acidobacteriota
Actinomycetota
Bdellovibrionota
Thermodesulfobacteriota
Halobacteriota
Thermoproteota
Verrucomicrobiota
Patescibacteria
Cyanobacteriota
Chloroflexota
Deinococcota
Fusobacteriota
Planctomycetota
Gemmatimonadota
Nitrospirota
Methylomirabilota
Abditibacteriota
Armatimonadota
Campylobacterota
Other
B
Aerobic_type
Forms_Biofilms
Gram_type
Phylum
Aerobic type
Aerobic (48.1%)
Anaerobic (32.2%)
Facultatively Anaerobic (19.7%)
Tree scale: 0.1
Forms Biofilms
FALSE (44.1%)
TRUE (55.9%)
Unknown
Gram type
Negative (58.2%)
Positive (41.8%)
Top 10 Genus
Pseudomonas 2.02%
Lachnospiraceae 1.75%
Sphingomonas 1.51%
Paenibacillus 1.42%
Bacteroides 1.41%
Bacillus 1.18%
Blautia 1.06%
Massilia 1.03%
Flavobacterium 0.98%
Faecalibacterium 0.89%
C
n_hostg / n_hosts (host count)
n_hostg (host genera)
n_hosts (host species)
log10(mean rel. abund.)
Pseudomonas
Pantoea
Sphingomonas
Bacillus
Enterobacter
Escherichia−Shigella
Paenibacillus
Massilia
Acinetobacter
Methylobacterium
Klebsiella
Citrobacter
Acidovorax
Agrobacterium
Erwinia
Stenotrophomonas
Rhizobium
Streptomyces
Chryseobacterium
Curtobacterium
Brevundimonas
Devosia
Kosakonia
Novosphingobium
Aquabacterium
D
n_hostg (host genera)
n_hosts (host species)
log10(mean rel. abund.)
mean relative abundance (log10)
Aspergillus
Alternaria
Fusarium
Davidiella
Penicillium
Filobasidium
Vishniacozyma
Malassezia
Aureobasidium
Acremonium
Diaporthe
Cercospora
Colletotrichum
Cystofilobasidium
Sporobolomyces
Stemphylium
Ramularia
Papiliotrema
Candida
Wallemia
Cladosporium
Curvularia
Itersonilia
Neoascochyta
Orbilia

Figure 4 Phylogenetic distribution, functional traits and host breadth of bacterial and fungal generalists and widespread genera. (A) Maximum-likelihood phylogeny of 1,103 bacterial generalist ASVs (16S; *n_hosts* ≥ 5 and not significantly host-clustered), with colour strips indicating phylum/class and BugBase traits, outer bars showing the number of host species harbouring each ASV, and an inset showing the 10 most frequently represented genera. (B) As in A for 5,830 bacterial specialist ASVs (*n_hosts* = 1), with the 10 most frequently represented genera shown in the inset. (C) The 25 bacterial genera spanning the largest number of host genera, with bars showing the number of host genera (*n_hostg*), host species (*n_hosts*) and mean relative abundance (log10). (D) As in C for the 25 most widespread fungal genera identified from ITS data; sequences unassigned at the genus level are excluded.

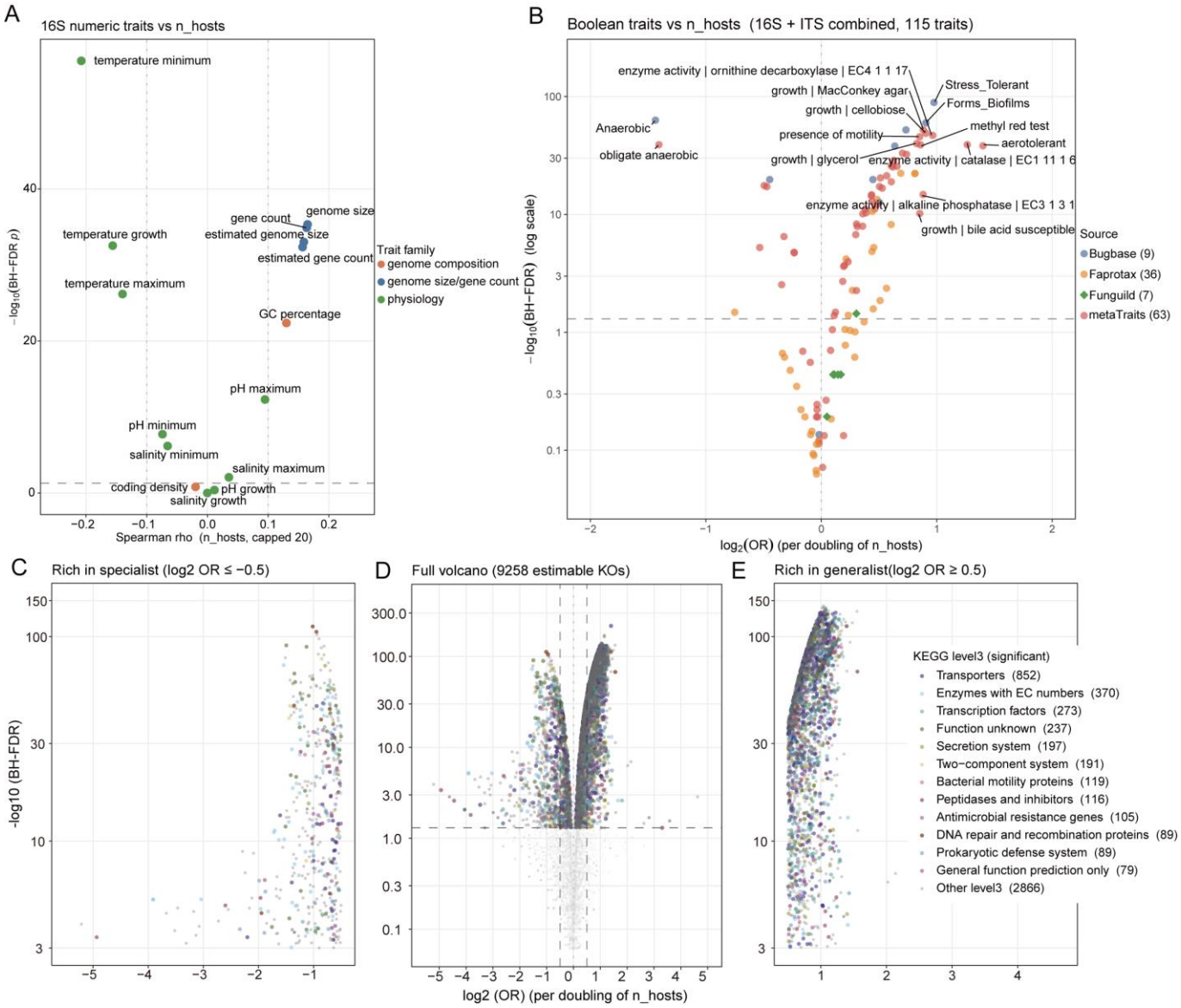


Figure 5 Generalists and specialists of the seed microbiome differ systematically in predicted functional traits. (A) Volcano plot of 15 continuous traits (metaTraits, 16S) associated with host breadth (*n_hosts*), with significant associations highlighted (BH-FDR < 0.05). (B) Volcano plot of Boolean traits from BugBase, FAPROTAX, metaTraits (16S) and FUNGuild (ITS); significant associations are highlighted (FDR < 0.05). (C) Specialist-enriched KO families (746; log2 OR ≤ −0.5), grouped by functional category. (D) Volcano plot of 9,580 KO families across 9,994 bacterial ASVs; 322 near-separated families with undefined ORs are omitted, and significant associations with host breadth are highlighted. (E) Functional categories of the 2,781 generalist-enriched KO families (log2 OR ≥ 0.5), with their relative proportions shown.

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Declaration of interests

The authors declare no competing interests.